

TUSHAR JAJU

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SUMMARY

ML Engineer with published research in cross-modal LLM adaptation (Springer, IMPACT 2025). Fine-tuned Mistral-7B with a 5-method curriculum framework achieving 58% video summarization improvement while preventing catastrophic forgetting. Built production pipelines on consumer GPUs: LoRA fine-tuning of 4B–7B models, custom datasets (4,300+ samples), and 4-bit quantized inference under 16GB VRAM.

PUBLICATIONS & OPEN SOURCE

- T. Jaju et al., “Cross-Modal Transfer Learning in Domain-Adaptive Video Summarization,” *IMPACT 2025*, Springer (presented Dec 6, 2025; proceedings forthcoming)
- Fine-tuned models and adapters published on [HuggingFace](#) (MedScribe SOAP LoRA, Bawarchi recipe generator); curated datasets on [Kaggle](#) (712 clinical samples, 68MB recipe corpus)
- [llmclean](#) – zero-dependency Python library for cleaning and normalizing raw LLM output (PyPI); [GitHub](#)
- Submission to Google MedGemma Impact Challenge 2026 (Main Track + Novel Task Prize)
- Ranked 78th out of 1,100 applicants in AnaVerse 2.0, national hackathon hosted by Celebal Technologies

PROJECTS

Hybrid-Dataset Summariser – Cross-Modal LLM Adaptation

Nov 2025 – Present

Tech: PyTorch (nightly/cu128), Mistral-7B, PEFT, bitsandbytes, HDF5, SBERT, Whisper | [GitHub](#)

- **Problem & data:** Published finding (IMPACT 2025) that LoRA fine-tuning on academic papers degrades video summarization by 14–50%; curated 4,324-sample cross-modal dataset (2,368 arXiv papers + 738 YouTube lectures + 1,218 SBERT-mined pairs) with full pipeline: yt-dlp scraping, Whisper transcription, GPT-4o-mini summaries, SBERT pair mining, quality filtering, Mistral tokenization, HDF5 packaging
- **Framework:** 3-phase curriculum on RTX 5070 Ti (16GB): LoRA+ asymmetric LR ($\eta_B/\eta_A=8$), OPLoRA orthogonal projection ($k=16$), EWC Fisher regularization ($\lambda=200\rightarrow400$), contrastive alignment; 4-bit NF4 + gradient checkpointing + 8-bit AdamW
- **Results:** Video ROUGE-1 +58% (0.263 $\rightarrow$ 0.417), video ROUGE-2 +272%, passive voice contamination held at 14.1% vs. 31.4% catastrophic baseline, cross-modal consistency +49%, BERTScore flipped negative $\rightarrow$ positive; all results statistically significant ($p<0.005$, paired t-test with bootstrap CI)
- **Evaluation:** 1,400-line evaluation harness: ROUGE-1/2/L, BERTScore, passive voice %, nominalization %, type-token ratio, D_gap composite metric with phase-gating, paired t-test, Cohen’s d, Wilcoxon signed-rank, bootstrap 95% CI

MedScribe – Clinical Documentation Workstation

Jan 2026 – Feb 2026

Tech: MedGemma 1.5 4B, MedASR Conformer 105M, LoRA, Gradio, CUDA | [GitHub](#) [HF](#) [Kaggle](#)

- **Pipeline:** MedASR (105M Conformer, 5.2% WER) $\rightarrow$ MedGemma 1.5 4B + LoRA $\rightarrow$ structured SOAP note + 6 clinical inference tools; total VRAM 3.4GB via 4-bit NF4 double quantization
- **Fine-tuning:** LoRA rank-16 on 712 GPT-4o curated samples (\$1.28 total cost); val loss 0.782 < train loss 0.828 (no overfitting); custom SOAPCompletionCriteria + StopOnRepetition guard eliminated 12% repetition failure mode
- **Impact:** 46% shorter notes vs. base MedGemma, 0% hallucinated findings, 100% section completeness (S/O/A/P); estimated 11 min/encounter saved; fully offline (no patient data egress)

Bawarchi – Role-Aware Recipe Generation with Cuisine-Context Switching

Nov 2025 – Jan 2026

Tech: YOLOv8, Llama 3.2 3B, Sentence-Transformers, Streamlit, CUDA | [GitHub](#) [HF](#) [Kaggle](#)

- **3-model pipeline:** YOLOv8m ingredient detection (7,431 images, 66.5% mAP, 202 manually annotated Indian spice images) $\rightarrow$ PMI + embedding substitution engine (522K recipes, 85% precision) $\rightarrow$ Llama 3.2 3B LoRA fine-tuning (335K recipes, 74.1% accuracy)
- **Innovation:** Cuisine-context switching generates culturally appropriate recipes (tacos vs. roti wrap) from identical ingredient sets; role-aware cross-cuisine protein substitution (chicken $\leftrightarrow$ paneer); published 68MB Kaggle dataset

EXPERIENCE

Data Science Intern – Vinkura AI

Aug 2025 – Dec 2025

- Built predictive, automated pipelines on 100K+ records; improved F1-score from 0.72 to 0.83 (15% gain)

TECHNICAL SKILLS

- **LLM Fine-tuning:** LoRA, EWC, PEFT, curriculum learning, Quantization, gradient checkpointing
- **Frameworks:** PyTorch, Transformers, bitsandbytes.
- **Data & Deployment:** HuggingFace Hub, Kaggle, Gradio, Streamlit, Git, Python, SQL

EDUCATION

B.Tech, Computer Science and Engineering

ABES Engineering College, Ghaziabad (AKTU)